Jenkins

Page No :

Jenkins Roles and Responsibilitys :-

Jenkins side we will create first CI Job and also we will perform the daily Releases, weekly Releases and monthly Releases and also we will create the Jobs multiple Jobs ~~and~~ also like pipeline Jobs, Freestyle Jobs. Copy existing Jobs and also Integration Automation process with Nexus Repository, code quality and also deploy the images Jenkins to Aws cloud we can deploy this types Roles and Responsibles we will perform.

Explain Jenkins work flow :

The work flow of Jenkins we are following like 7 stages we configure inside the Job i.e first stage is code validation once it is complete then second stage build stage Third stage is code quality, fourth phase Release phase Fivth stage is uploding the package into server i.e RHEV serve sixth stage is Deployment, seventh stage is testing. everything is automated in single Job. How to automated is for this we are ~~automated~~ enable the pollscm inside the Job what exactly pollscm will do is whenever developer checked in the code in versione control system My code in Jenkins Automatically it will always kept the code like whenver changes happen in version control system it will pull the That changes and then it will create the package i mean whenever it will start build my sonarqube enable it will check the code quality Basically we are using the code quality purpose sonarqube. Here we have a different different stages ie critical, major, minor, blocker Basically whenever critical happen it will stop the build process can't go the Next level. Suppose there is No issue it will in The code Then it will go to the release stage. Once it is Completed Then package will be uploded into The Redhat Satilite server Finally Rundeck we using for the Deploy The package into different different environment staging. and also UAT, QA So That IP Address we are passing from the Jenkins Job only

What Kind of ~~java~~ Applications you are using in Jenkins

An Java, oracle Applications

What are the Requirements of Jenkins?

→ Build tool maven.
→ Java Application
→ Version control System
→ Deployment tool, Rundeck.

Scanned by CamScanner

* How to Create a Slave Nodes?

Coming to master slave consept in Jenkins what we have to do is main purpose of master slave purpose is to distributing the load into different different slave machines usevally slave machine how it will create is I will assigne the separate Ip address for the perticlur slave so in the perticlur slave what I will do is I will Install the whatever Requirement I have in the manage Jenkins I will click on the manage Nodes Their I will click New Node for the Requirement if I request slave1 Name I will give the name slave 1 Then ok it will show the configure dash board Here I will give the Remote Root fs path so what is the main purpose of Remote Root fs is whatever doing The Jenkins loks in frentend backed that locks stored that thing I will give Remote Root fs path. later took Location I will Check Took location is Nothing but exactly where we want to call I mean suppose JAVA 1.7 I want to call in the Jenkins machine I will give The That perticlur path and also maven path I should give and also ANT path I should give. whatever requirement I have That will pass from that location soo after that I launch The slave machine in the jenkins. later once I create slave machine ~~The Jenkins~~ Then I can use That Jenkins job.

* Without Continous Intergration what will happen?

So with out continous Integration what happend is so Basically currently working on something development so you are doing the development something 1.0 version whatever it may be with out Continous Intigratin what will do is

* Deployment pipeline in Jenkins?

Deployment pipeline we create for the stage enviroment and UIT enviroment pipeline view. Basically we are using the Build pipeline plugin for that so Respected Job's First we need to configure that staging environment Job and UIT environment so that Job configure once It is completed which Job I need to check whether the builds are happing are not I mean I gave configuration is correct or not once It is completed Then we need to attach the Job's into pipeline process so pipeline process. So we are configure pipeline plugin I mean Build pipe line plugin so Their in first Job I need to check ~~in~~ The intial Job from where which enviroment we need to start ~~peple~~ building for pipeline view so that we can give the conecting b/w Jobs using the

* Continous Delivery:-

deploying the product to customers continously with ~~without~~ Approval. This is not Automatic

Continous deployment:-

~~Continous~~ Deploying the product to customers continously without Approval This is automatic.

Nexus

Nexus basically using for backup for each & every release version i mean packages That packages i can store in nexus repository

* Even Though after one year (or) two years if we want to get back old version packages we can get back from The Nexus.
* Repository That purpose we can integrated with Jenkins
* Basically from maven only we can integrated nexus.
* No Need to take backup separtly it will take automatica
* Whenever we checked into The Code into The version control system
* I Can integrated nexus inside The maven only in The maven I can write ~~destribution~~ mgnt in The pom.xml
* Accoding to That it will take The Code and it will uploded into The Nexus repository That senario we are using in my environment it will go to Nexus.

* How to take backup in Jenkins?

* Comming to the backup in Jenkins so we are taking the backup usually Nexus repository we will use. Jenkins only.
* Inside the Jenkins I will configure the maven so in maven we have 3rd party repository
* for each & every Release i will take the backup of my packages in your Nexus server that configuration I will give in the maven settings.xml
* I will configure the Nexus that perticular URL. The repository path i will set in the maven pom.xml using the distribution mngt tag i will setup the Nexus server.

* There is only issue in production environment how u can solve
* usually i will support only test environment but i have knowledge on production so i will explain.
* usually in production deployment what we will do is whenever i deploy the packages into production first
* I will prepare the implementation plans for the deployment suppose why because whenever i am doing activity major activity in the production.
* I will take the Approval from ~~the~~ my client if anything goes wrong in the production before i started work in the production i will take the snapshot of the perticular version why because every time whenever i do upgration don't know my upgration success (or) not That's why First i will take the snapshot perticular version.
* Later i will start deployment whatever it may be in production server suppose any-thing goes to wrong what i will do is i will roback the production server.
* if no issue in production server Then i will proceed with next release like that i will resolve problems.

* One machine is not running (or) one service is not runing what you will do first?

→ In my project we have helth check monitoring.

→ It is everyday it will check the servers & complete information about the servers which machine having more memory takes & all it will send into my Email ID.

* Before am doing anything production servers I will check ~~[illegible]~~ health monitoring list" it will shows which machine having more memory taken.

* Suppose if any machine taking more memory, I will remove my old logs into my particular machine then I will proceed with the production servers.

* This stategy I will use in my organization.

* How you are doing performance monitoring?

→ we are using "performance checkup".

* Who will maintain nagios server?

My client will maintain that server what I have to do as a devops engineer I supported from the off-shore client they have on-shore they will take the administration on nagios everything. They will take care I will maintain my servers in off-shore.

* Which type of webserver you are using?

Redhat Satilite server.

Jenkins Roles and Responsibilites:-

Jenkins Side we will create first CI Job and also we will perform the Daily Rules Jobs, monthly Releses and also we will Create the Jobs multiple Jobs like pipeline Jobs Freestyle Jobs, Copy existing Jobs and also integration Automation process with, Nexus Repository, code quality or deploy and also we will do the deploy the images Jenkins to Aws cloud we can do deploy. This Types Roles and Responsible we will perform.

Flow of Jenkins:-

The work-flow Jenkins we are following like 7 stages we configure inside the Job. i.e Code Check in ~~Commit~~ is completed second stage is build stage, Third stage is Code quality fourth stage is Release phase. fifth is uploading package into the serves Rhel server, sixth stages is deployment Seventh stage is Testing. every thing we Automated Job in single Job How to Automated is for this we are enable the pollscm inside the Job what exactly pollscm will do is whenever you Checkd in the code we will do the version control system automatically my code in the Jenkins Automatically it will always kept the code like whenever Changes happen in vcs it will pull the That Changes and then it will create the package i mean whenever it will start build my sonar qube enable my sonarqube scanner will be enable it will Check the code quality Basically we are using the code quality purpose sonar qube Here we have a diferent diferent stages Critical, major, minor, block Basicaly whenever Critical happen it will stop the build process it won't go to Next level. Suppose there No issue in code then it will go to relese stage. once it is completed The package will be upload in The Redhat catilite Server Finaly Rundeck we using for the Deploy The package into The different deffent envirolment Staging, and also UID, QA so That Ip Address we are passing from The Jenkins Job only The build is perameterised Thus we are passing The perameter Ip Address where we want to deploy your package. once it is completed That configuration part after That I check in The code and Then step by step I will execute The Jenkin Job Finally I need login to That perticlug server deployment happen or not so These workflow we are follow.

→ What kind of Applications you are using in Jenkins
Ans Java Applications.

→ What are the Requirements of Jenkins
* MAven build tool we required Java Application and version control system ⇒ CI tool is required and deployment is required.

New members'
you are asked what he told you
→ what are the steps you will perform ~~the~~ steps give the access on Jenkins?

→ Basically we are ~~passing the~~ passing access in two ways i mean specific Job usually we are using for this Authentication for the development team if he is the guy is my team ~~team~~ member then i will go for the manage Jenkins → ~~configuration~~ ~~option~~ security global → then I should select the project specific access there i can add new member Id and then I can give the administration access and also configure update and ~~apps~~ plugin installation every thing i will give and I save the configuration. after that I tell him to login using credentials or whatever he have credentials and after that I ask him to create one dummy e2 Job try to build the job. This is the way I will proceed.

→ what ~~are~~ you have done in Chef and what are the things you have done in chef?

Chef I worked many times doing the deployment to the environment for that I written the cookbooks in the chef and also I implemented chef infrastructure in my environment completely end to end process Administration I worked I mean installing the chef server and chef manager chef workstation and chef client. I configured and also i configured the cook books according to the requirements for upgration and deploy different differ environments so this kind of activity I worked.

* How you can move or copy jenkins from one server to another?

* Slide a Job from installation of Jenkins to another by copying the related job directory.
* make a copy of an already existing job by making clone a Job directory by a different name.
* Renaming an existing job by renaming a directry.

* ~~Change management~~ :-

ITSM ⇒ tool we are using Rise a request

Whenever we are doing the activity in our Jenkins or Server side we are Riseing a change ~~ment~~ Request and prepareing The implimention plans in The Change Mnt Tool.

* Customized Resource in Chef?

Scanned by CamScanner